

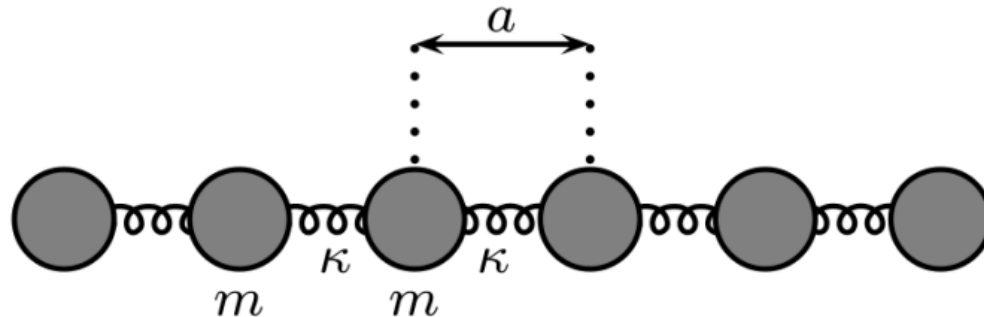
Physical Chemistry of Materials

Lecture 12

Phonons



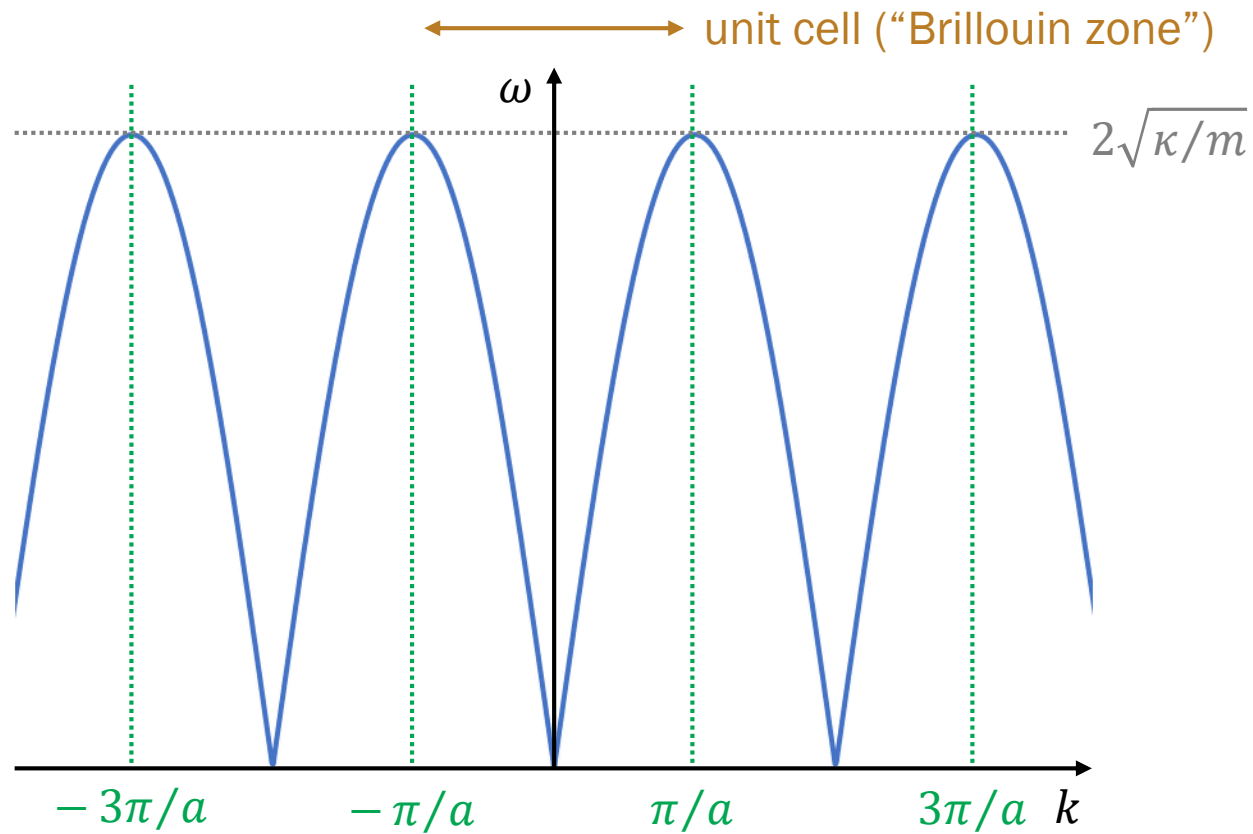
Normal Modes of 1D Crystal



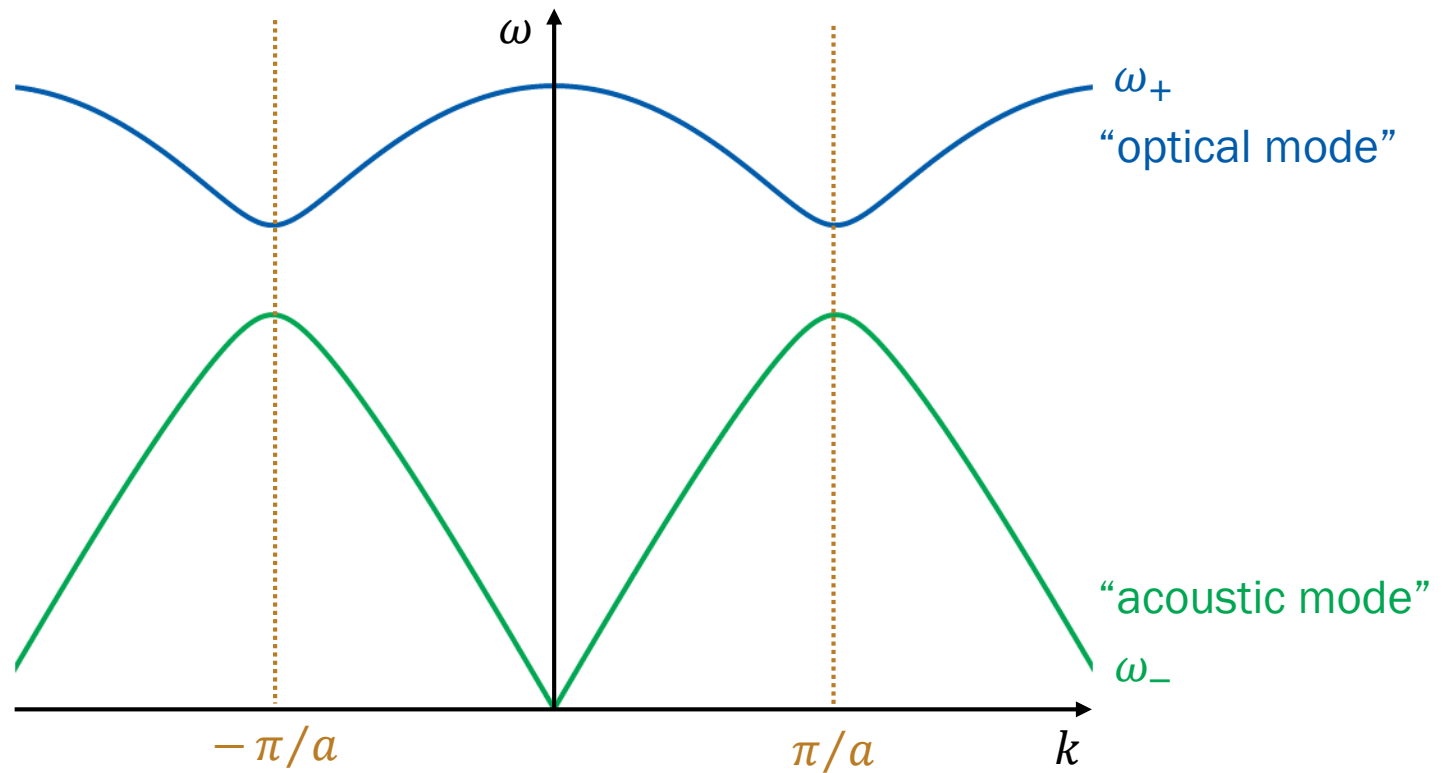
- Deviation from equilibrium position $q_n = x_n - x_{n0}$
- Total potential energy

$$V_{\text{tot}} = \sum_{i=1}^N \frac{\kappa}{2} (x_{i+1} - x_i - a)^2 = \sum_{i=1}^N \frac{\kappa}{2} (q_{i+1} - q_i)^2$$

Dispersion Relation

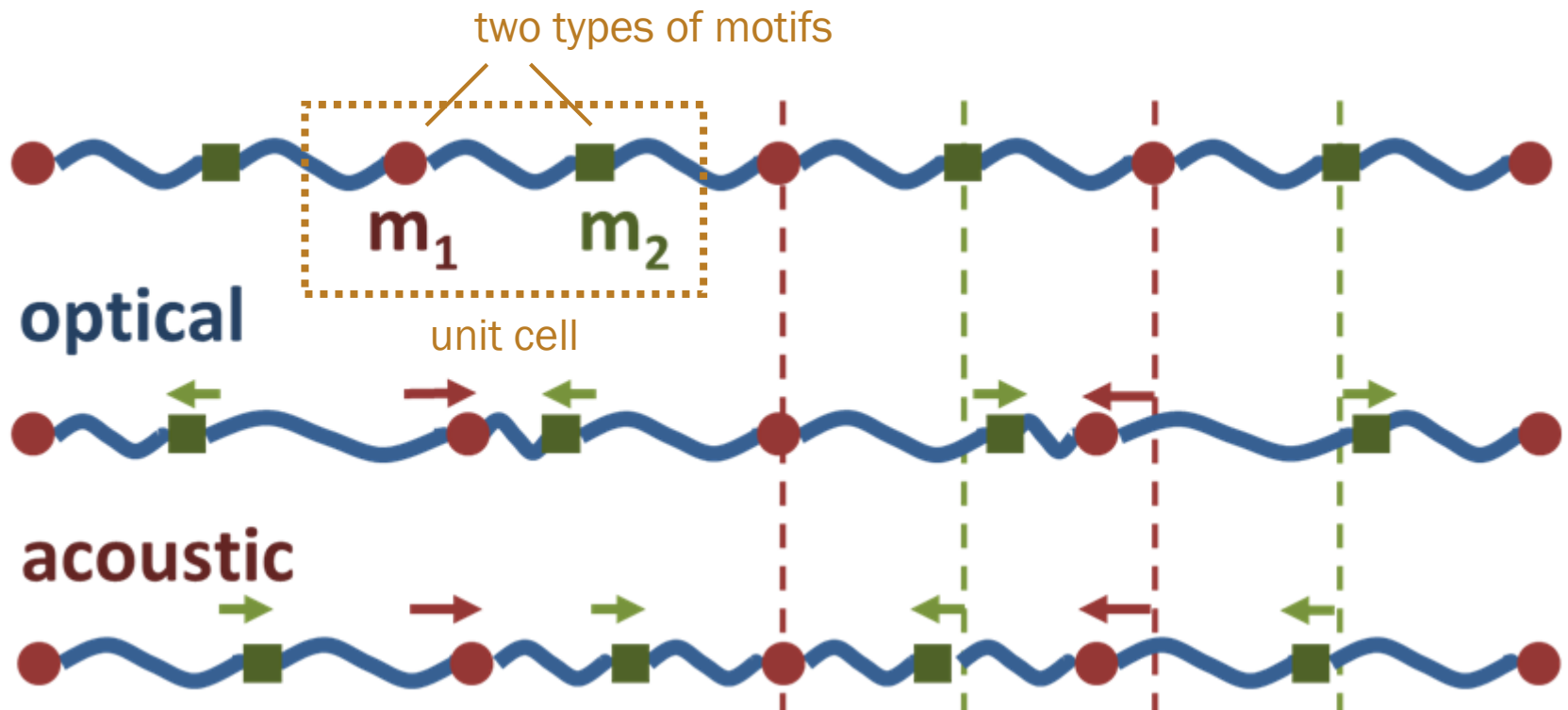


Case of Diatomic Solid



Last Class

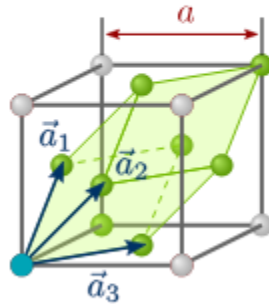
Acoustic and Optical Phonons



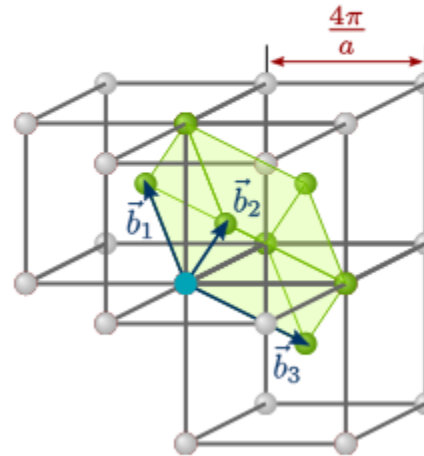
(Image: https://commons.wikimedia.org/wiki/File:Optical_%26_acoustic_vibrations.png)

Reciprocal Lattice in 3D

direct lattice:
fcc with edge length a



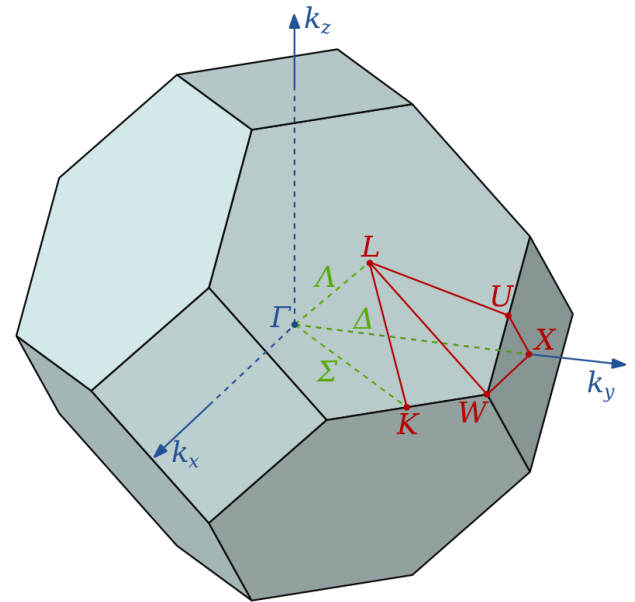
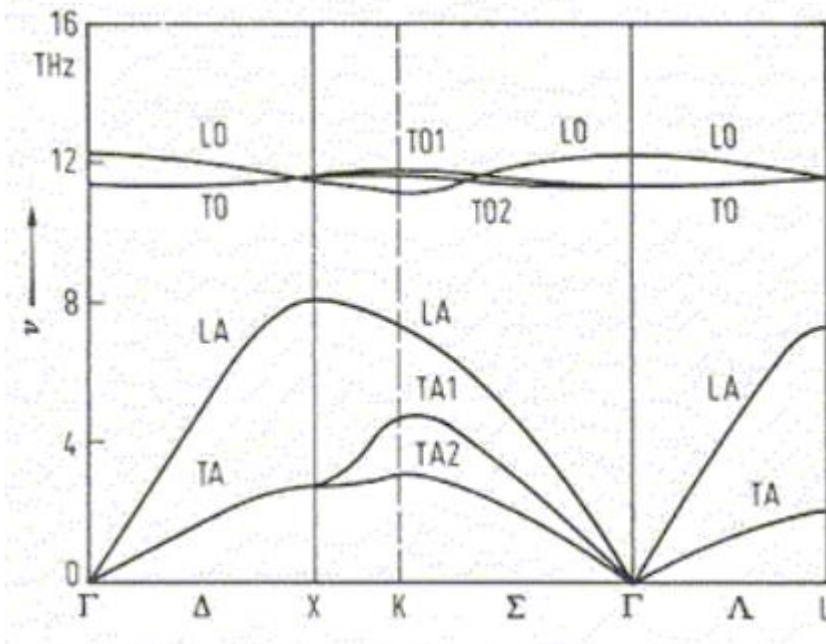
reciprocal lattice:
bcc with edge length $4\pi/a$



$$\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{ij}$$

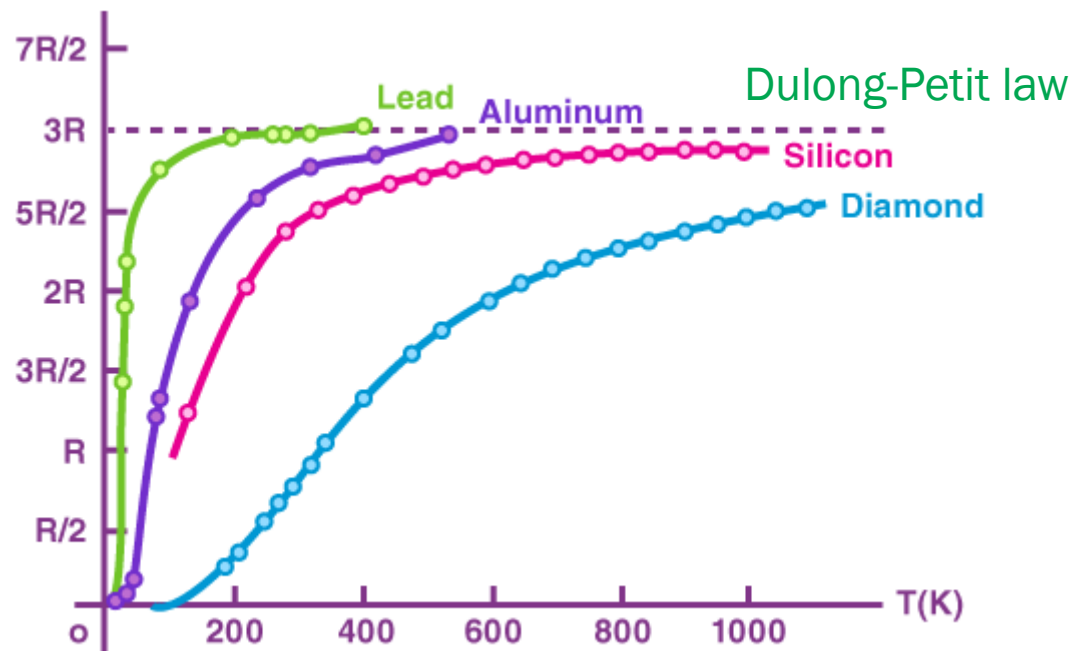
(Image: <https://www.physics-in-a-nutshell.com/article/15/the-reciprocal-lattice>)

Dispersion Relation in 3D



Heat Capacity of Solid

molar heat capacity



(Image: <https://blog.naver.com/jinest/70153707595>)

Classical Treatment

- $3N$ independent (identical) harmonic oscillators

$$Z = z_{\text{vib}}^{3N}$$

$$z_{\text{vib}} = \frac{1}{h} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\beta \left(\frac{1}{2} k x^2 + \frac{p^2}{2m} \right)} dx dp = \frac{k_B T}{h\nu}$$

$$Z = \left(\frac{k_B T}{h\nu} \right)^{3N} = (\beta h\nu)^{-3N}$$

Heat Capacity

$$Z = (\beta h\nu)^{-3N}$$

$$U = -\frac{\partial \ln Z}{\partial \beta} = \frac{3N}{\beta} = 3Nk_{\text{B}}T = 3nRT$$

Hence, the heat capacity is

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V = 3nR$$

(Dulong-Petit law)

Improvement: Normal Modes

- $3N$ independent (different) harmonic oscillators

$$Z_{\text{vib},i} = \frac{k_B T}{h \nu_i}$$

$$Z = Z_{\text{vib},1} Z_{\text{vib},2} \cdots Z_{\text{vib},3N} = \prod_{i=1}^{3N} Z_{\text{vib},i}$$

$$Z = \left(\frac{k_B T}{h} \right)^{3N} \frac{1}{\prod_{i=1}^{3N} \nu_i} = (\beta h)^{-3N} \frac{1}{\prod_{i=1}^{3N} \nu_i}$$

Heat Capacity

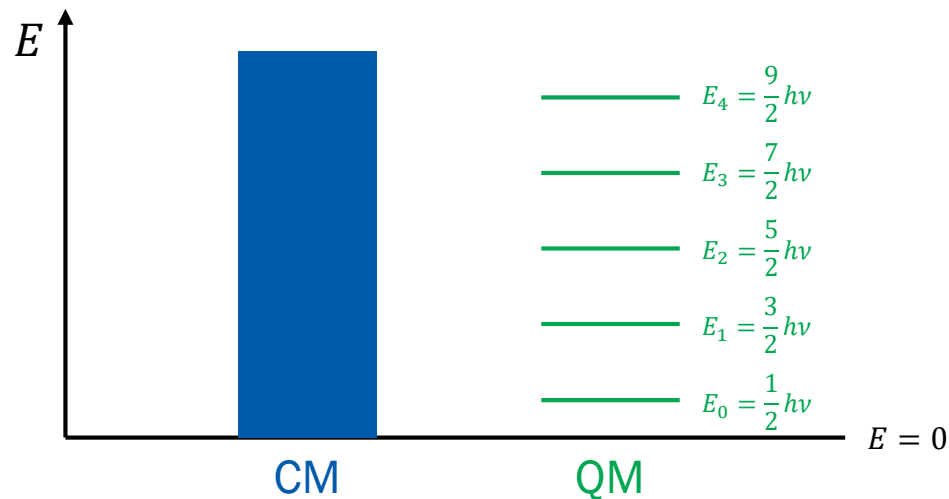
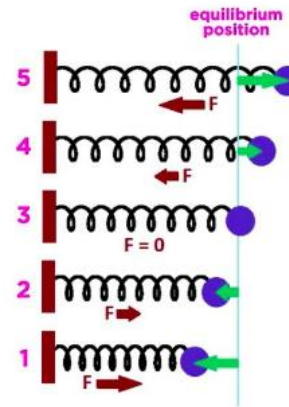
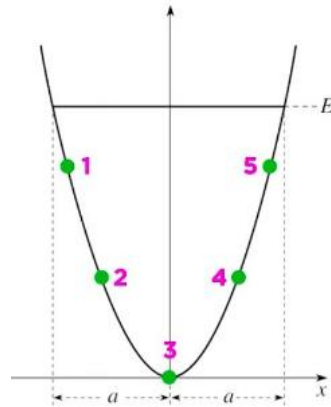
$$Z = (\beta h)^{-3N} \frac{1}{\prod_{i=1}^{3N} v_i}$$

$$U = -\frac{\partial \ln Z}{\partial \beta} = \frac{3N}{\beta} = 3Nk_B T = 3nRT$$

Hence, the heat capacity is (again)

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V = 3nR$$

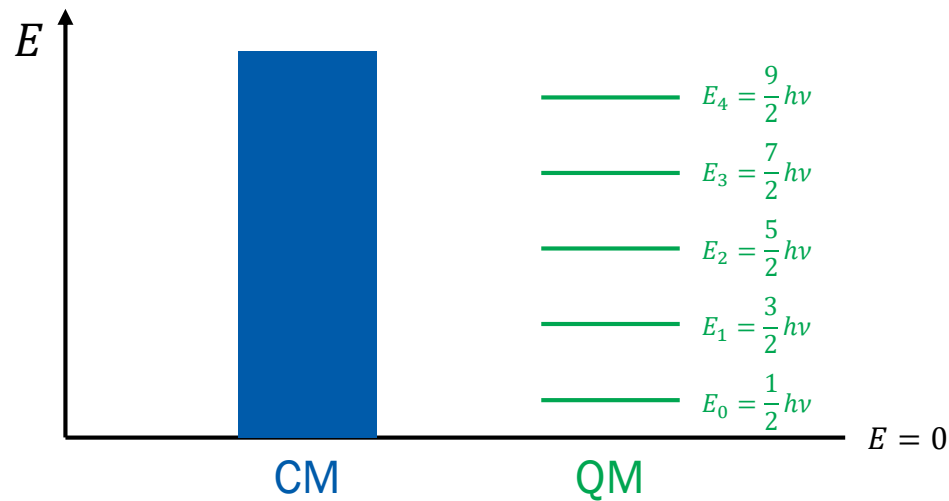
Quantization of Vibration



(Image: https://www.youtube.com/watch?v=yG_Ot9rsNaw)

“Phonon”

A **phonon** is a discrete **quantum of vibration**
(bosonic)



Einstein Model

- $3N$ independent (identical) harmonic oscillators

$$Z = z_{\text{vib,QM}}^{3N}$$

Note: $\omega = 2\pi\nu$

$$z_{\text{vib,QM}} = \sum_{v=0}^{\infty} e^{-\beta\hbar\omega\left(v+\frac{1}{2}\right)} = \frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}}$$

$$Z = \left(\frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}} \right)^{3N}$$

Heat Capacity

$$Z = \left(\frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}} \right)^{3N}$$

$$U = -\frac{\partial \ln Z}{\partial \beta} = \frac{\partial}{\partial \beta} \left[\frac{3}{2} N\hbar\omega\beta + 3N \ln(1 - e^{-\beta\hbar\omega}) \right]$$

$$U = \frac{3}{2} N\hbar\omega + 3N\hbar\omega \frac{e^{-\beta\hbar\omega}}{1 - e^{-\beta\hbar\omega}} = \frac{3}{2} N\hbar\omega + 3N\hbar\omega \frac{1}{e^{\beta\hbar\omega} - 1}$$

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V = \frac{d\beta}{dT} \frac{\partial U}{\partial \beta} = \frac{1}{k_B T^2} \cdot \frac{3N(\hbar\omega)^2 e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega} - 1)^2}$$

Einstein's Prediction

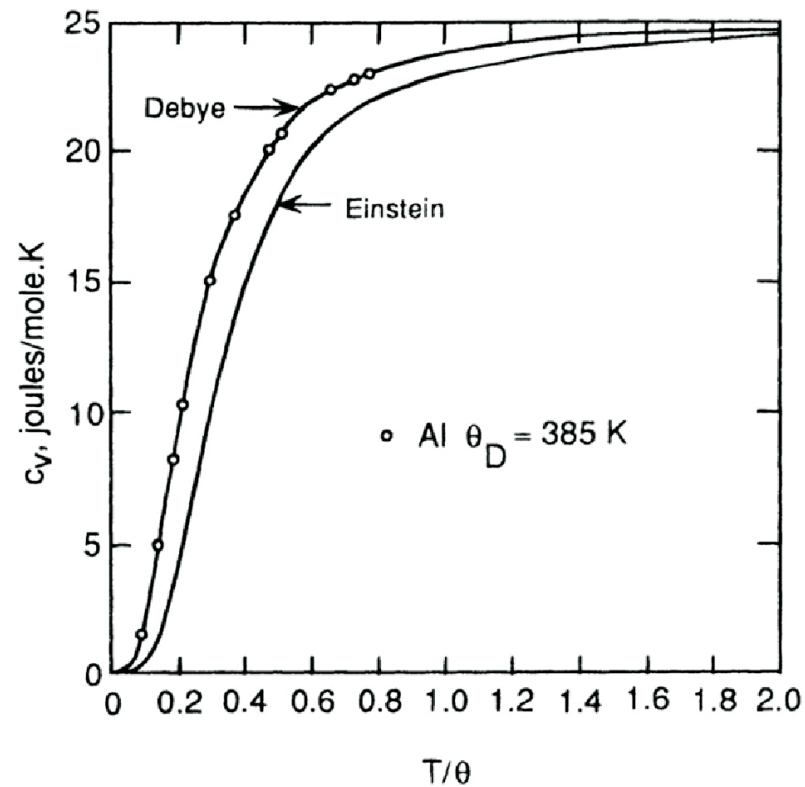
$$C_V = \frac{1}{k_B T^2} \cdot \frac{3N(\hbar\omega)^2 e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega} - 1)^2}$$

$$C_V = 3Nk_B(\beta\hbar\omega)^2 \frac{e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega} - 1)^2}$$

As $T \rightarrow 0$, $e^{\beta\hbar\omega} = e^{\hbar\omega/k_B T} \gg 1$ and

$$C_V \approx 3Nk_B(\beta\hbar\omega)^2 e^{-\beta\hbar\omega} \propto \frac{1}{T^2} e^{-\hbar\omega/k_B T}$$

Experimental Data



(Image: https://www.tf.uni-kiel.de/matwis/amat/td_kin_i/kap_1/backbone/r_su4.html)

Boson Counting

$$U = \frac{3}{2}N\hbar\omega + 3N\hbar\omega \frac{1}{e^{\beta\hbar\omega} - 1}$$

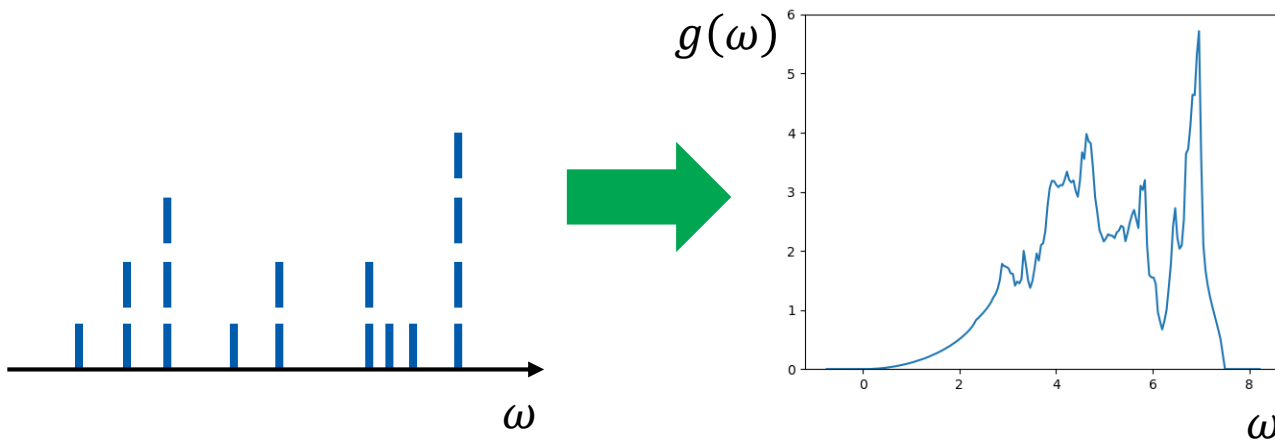
Ignoring the zero-point energy (constant),

$$U = 3N\hbar\omega \frac{1}{e^{\beta\hbar\omega} - 1} = 3N\hbar\omega N_{\text{BE}}(\hbar\omega)$$

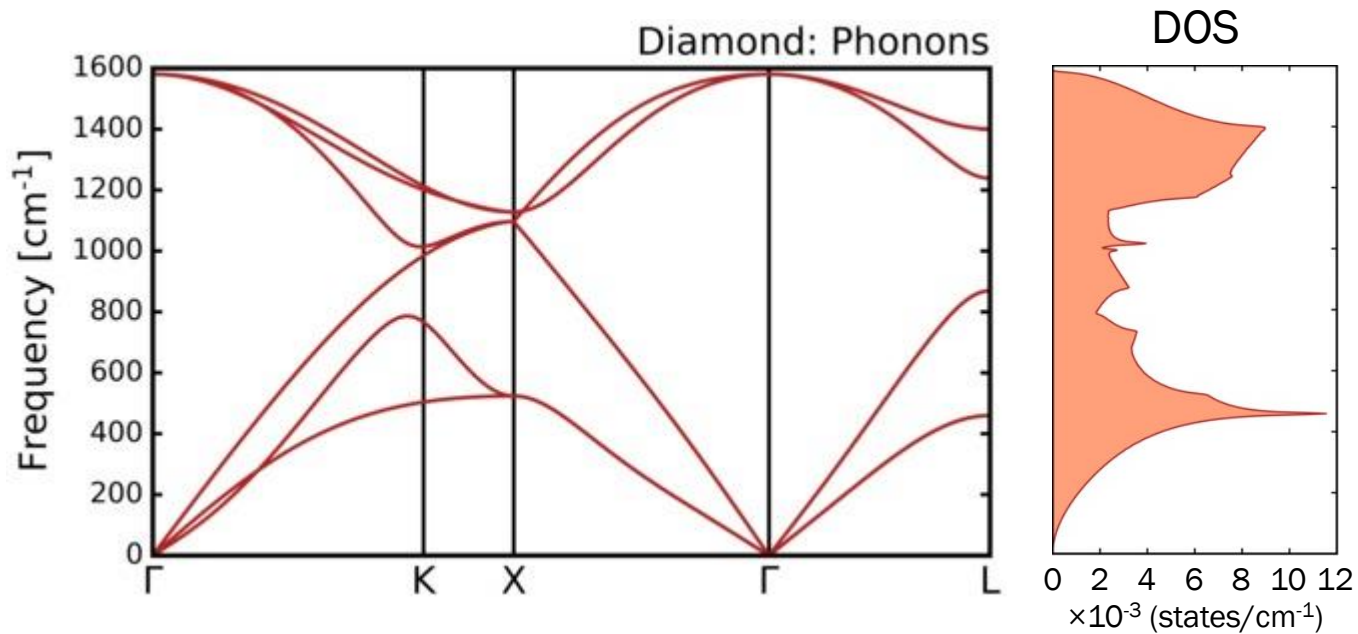
where $N_{\text{BE}}(\hbar\omega)$ is the average occupation number of bosons for a microstate with energy $\hbar\omega$

Debye Model

- $3N$ independent (different) harmonic oscillators
- We have large N : many normal modes with different ω_i
- Consider them as continuous!
- Density of states (DOS) $g(\omega)$



Determination of DOS



(Image: https://www.tf.uni-kiel.de/matwis/amat/td_kin_i/kap_1/backbone/r_su4.html)

DOS of Simple Model

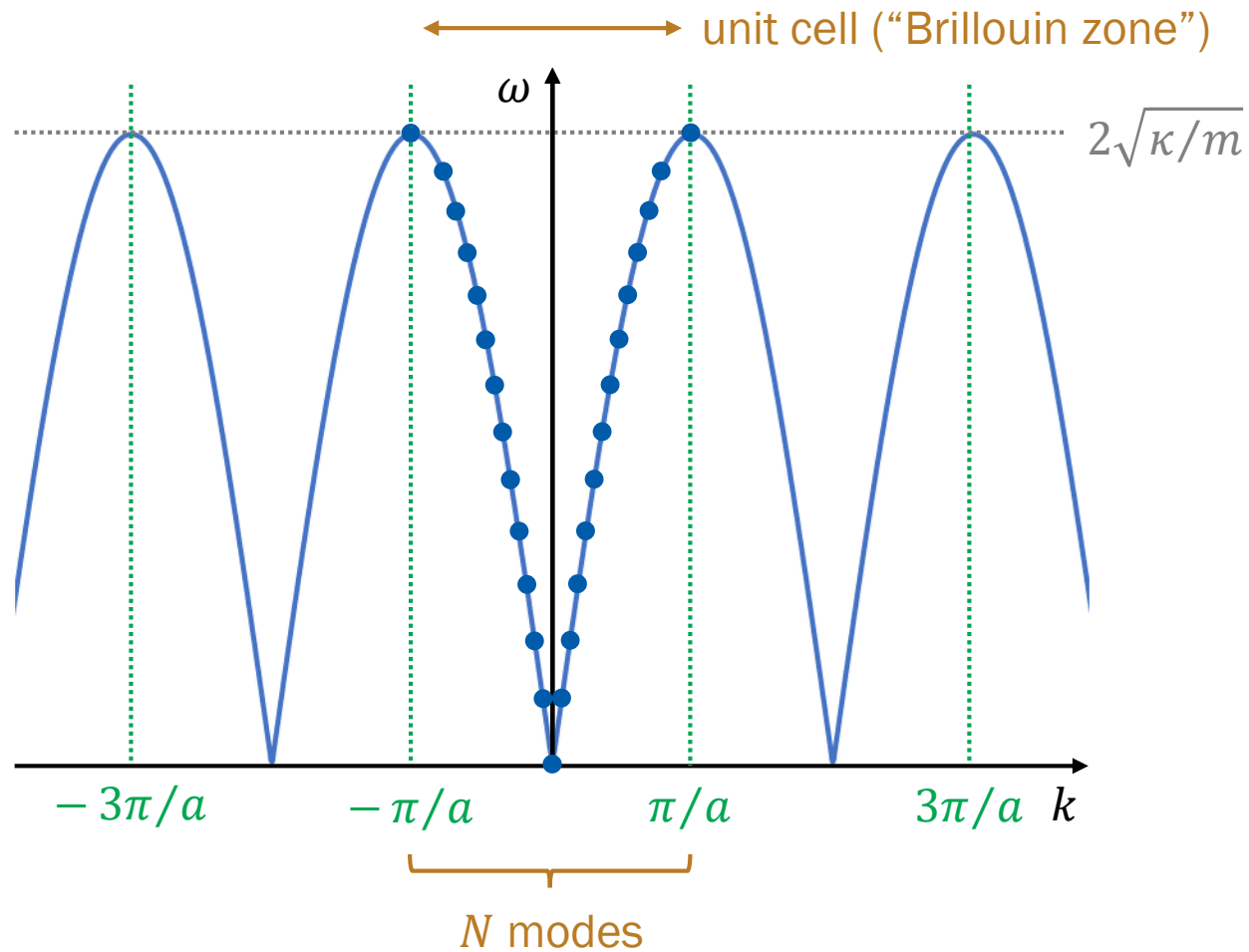
- $g(\omega) d\omega = dN_{\text{modes}}$: number of modes in $[\omega, \omega + d\omega]$
- $h(k) dk = dN_{\text{modes}}$: number of modes in $[k, k + dk]$

$$h(k) = \frac{dN_{\text{modes}}}{dk}$$

How can we count the number of modes?

The dispersion relation is continuous only when $N \rightarrow \infty$;
consider finite N (= finite system size)!

Case of 1D Simple Model



DOS Calculation

- $g(\omega) d\omega = dN_{\text{modes}}$: number of modes in $[\omega, \omega + d\omega]$
- $h(k) dk = dN_{\text{modes}}$: number of modes in $[k, k + dk]$

$$h(k) = \frac{dN_{\text{modes}}}{dk}$$

$$h_{1D}(k) \sim \frac{N}{2\pi/a} = \frac{L}{2\pi}$$

$$h_{3D}(\vec{k}) \sim \frac{N^3}{(2\pi/a)^3} = \frac{L^3}{(2\pi)^3} = \frac{V}{(2\pi)^3}$$

DOS Calculation

- $g(\omega) d\omega = dN_{\text{modes}}$: number of modes in $[\omega, \omega + d\omega]$
- $h(k) dk = dN_{\text{modes}}$: number of modes in $[k, k + dk]$

$$h_{3D}(\vec{k}) \sim \frac{N^3}{(2\pi/a)^3} = \frac{L^3}{(2\pi)^3} = \frac{V}{(2\pi)^3}$$

The number of modes in $[0, k]$ is then

$$N_{\text{modes}}(k) = \int_{\text{volume}} h_{3D}(\vec{k}) d\vec{k} = \int_0^k h_{3D}(\vec{k}) 4\pi k^2 dk$$

$$N_{\text{modes}}(k) = \frac{V k^3}{6\pi^2}$$

DOS Calculation

- $g(\omega) d\omega = dN_{\text{modes}}$: number of modes in $[\omega, \omega + d\omega]$

The number of modes in $[0, k]$ is

$$N_{\text{modes}}(k) = \frac{Vk^3}{6\pi^2}$$

$$g(\omega) = \frac{dN_{\text{modes}}}{d\omega} = \frac{Vk^2}{2\pi^2} \frac{1}{d\omega/dk}$$

Density of States

$$g(\omega) = \frac{V k^2}{2\pi^2} \frac{1}{d\omega/dk} \quad \text{Note: } \omega = 2\sqrt{\frac{\kappa}{m}} \left| \sin\left(\frac{ka}{2}\right) \right|$$

Assuming a simple dispersion relation, $\omega = vk$,

$$g(\omega) = \frac{V \omega^2}{2\pi^2 v^3}$$

Considering longitudinal and transverse waves,

$$g(\omega) = g_l(\omega) + 2g_t(\omega) = \frac{V \omega^2}{2\pi^2 v_l^3} + \frac{V \omega^2}{\pi^2 v_t^3}$$

Energy Calculation

$$g(\omega) = g_l(\omega) + 2g_t(\omega) = \frac{V\omega^2}{2\pi^2 v_l^3} + \frac{V\omega^2}{\pi^2 v_t^3}$$

$$U = \int_0^? g(\omega) \hbar \omega N_{\text{BE}}(\hbar \omega) d\omega$$

Do we have the upper limit?

Upper Limit ω_c

No waves can propagate inside the solid
with a too small wavelength (= too high frequency)

$$\int_0^{\omega_c} g(\omega) d\omega = 3N$$

$$\int_0^{\omega_c} \left(\frac{V}{2\pi^2 v_l^3} + \frac{V}{\pi^2 v_t^3} \right) \omega^2 d\omega = 3N$$

$$\omega_c^3 = 9N \left(\frac{V}{2\pi^2 v_l^3} + \frac{V}{\pi^2 v_t^3} \right)^{-1}$$

Energy Calculation

$$\omega_c^3 = 9N \left(\frac{V}{2\pi^2 v_l^3} + \frac{V}{\pi^2 v_t^3} \right)^{-1}$$

$$g(\omega) = \frac{V \omega^2}{2\pi^2 v_l^3} + \frac{V \omega^2}{\pi^2 v_t^3} = \frac{9N \omega^2}{\omega_c^3}$$

$$U = \int_0^{\omega_c} g(\omega) \hbar \omega N_{\text{BE}}(\hbar \omega) d\omega$$

$$U = \frac{9N \hbar}{\omega_c^3} \int_0^{\omega_c} \omega^3 \frac{1}{e^{\beta \hbar \omega} - 1} d\omega$$

(No closed form)

Heat Capacity

$$U = \frac{9N\hbar}{\omega_c^3} \int_0^{\omega_c} \omega^3 \frac{1}{e^{\beta\hbar\omega} - 1} d\omega$$

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V = \frac{d\beta}{dT} \frac{\partial U}{\partial \beta} = \frac{1}{k_B T^2} \frac{9N\hbar}{\omega_c^3} \int_0^{\omega_c} \frac{\hbar\omega^4 e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega} - 1)^2} d\omega$$

Letting $x = \beta\hbar\omega$ and $x_c = \beta\hbar\omega_c$,

$$C_V = \frac{9Nk_B}{x_c^3} \int_0^{x_c} \frac{x^4 e^x}{(e^x - 1)^2} dx$$

Debye T^3 Law

$$C_V = \frac{9Nk_B}{x_c^3} \int_0^{x_c} \frac{x^4 e^x}{(e^x - 1)^2} dx$$

As $T \rightarrow 0$, $x_c = \beta \hbar \omega_c \rightarrow \infty$, and

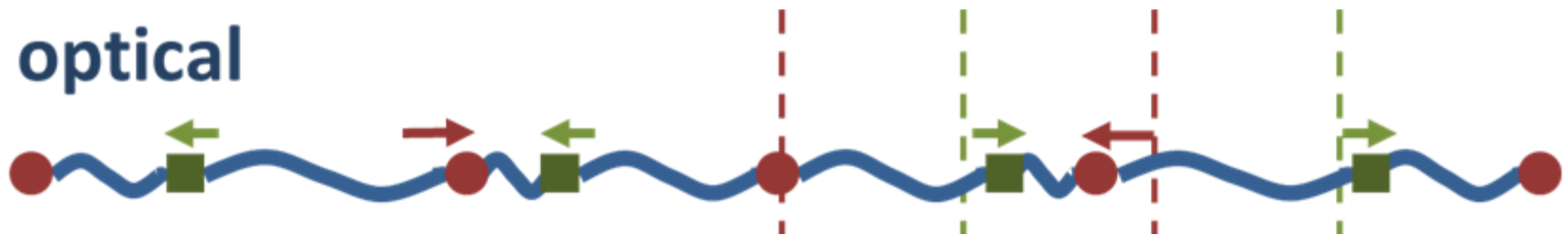
$$C_V \approx \frac{9Nk_B}{x_c^3} \int_0^{\infty} \frac{x^4 e^x}{(e^x - 1)^2} dx \propto \frac{1}{x_c^3} \propto \frac{1}{\beta^3} \propto T^3$$

Quantum: Selection Rules?

Vibrational Selection Rules

Dipole moment of the molecule must vary during a vibration

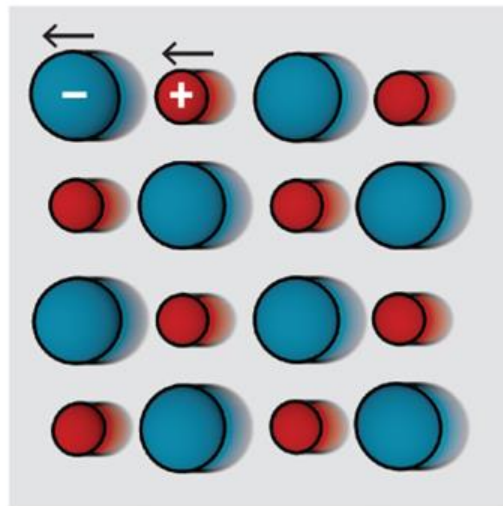
- Infrared active: the normal mode whose $d\mu/dq_j \neq 0$
- Infrared inactive: the normal mode whose $d\mu/dq_j = 0$



Optical phonons are IR active!

Two Types of Phonons

Acoustic

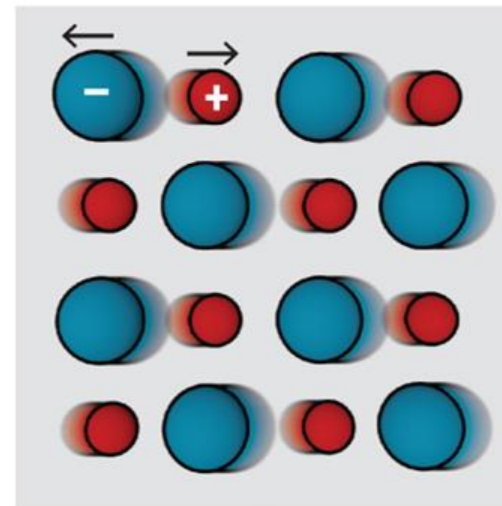


Heat



Sound

Optical



Light